

## School of Computer Science and Artificial Intelligence

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### Lab Assignment # 14

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#### **Task Description 1 – (Landing Page Design)**

• Create a simple landing page for a product or service. Include a header, hero section, and footer.

#### **Requirements:**

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.

#### **Prompt:-**

Create a simple landing page for a product or service website.

The page should include a header, a main content section, and a footer using proper HTML tags and structure.

#### **Code:-**

```
2 //Create a simple landing page for a product or service website.
3 The page should include a header, a main content section, and a footer using proper
4 HTML tags and structure.
5 <!DOCTYPE html>
6 <html lang="en">
7 <head>
8   <meta charset="UTF-8">
9   <meta name="viewport" content="width=device-width, initial-scale=1.0">
10  <title>Product Landing Page</title>
11  <style>
12    body {
13      font-family: Arial, sans-serif;
14      margin: 0;
15      padding: 0;
16    }
17    header {
18      background-color: #4CAF50;
19      color: white;
20      padding: 20px;
21      text-align: center;
22    }
23    main {
24      padding: 20px;
25    }
26    footer {
27      background-color: #333;
28      color: white;
29      text-align: center;
30      padding: 10px;
31      position: fixed;
32      width: 100%;
33      bottom: 0;
34    }
35  </style>
36 </head>
37 <body>
38   <header>
39     <h1>Welcome to Our Product</h1>
40     <p>Your solution for a better life</p>
41   </header>
42   <main>
```

```
36 </head>
37 <body>
38   <header>
39     <h1>Welcome to Our Product</h1>
40     <p>Your solution for a better life</p>
41   </header>
42   <main>
43     <h2>About Our Product</h2>
44     <p>Our product is designed to help you achieve your goals and improve your lifestyle. With cutting-edge technology and user
45     <h2>Features</h2>
46     <ul>
47       <li>Feature 1: Innovative design</li>
48       <li>Feature 2: Easy to use</li>
49       <li>Feature 3: Affordable price</li>
50     </ul>
51     <h2>Get Started</h2>
52     <p>Ready to experience the benefits of our product? Click the button below to get started!</p>
53     <button onclick="alert('Thank you for your interest!')">Get Started</button>
54   </main>
55   <footer>
56     &copy; 2024 Our Product. All rights reserved.
57   </footer>
58 </body>
59 </html>
60
```

## Output:-

//Create a simple landing page for a product or service website. The page should include a header, a main content section, and a footer using proper HTML tags and structure.

# Welcome to Our Product

Your solution for a better life

## About Our Product

Our product is designed to help you achieve your goals and improve your lifestyle. With cutting-edge technology and user-friendly design, it's the perfect choice for anyone looking to enhance their daily routine.

## Features

- Feature 1: Innovative design
- Feature 2: Easy to use
- Feature 3: Affordable price

## Get Started

Ready to experience the benefits of our product? Click the button below to get started!

Get Started

## Justification:-

Highlights a product section with clear headings and bullet points to explain important features and benefits.

Uses a simple and readable layout to help users understand the product easily and increase engagement.

## Task Description 2 – (Theme Toggle Feature)

- Provide a button to switch between light and dark themes..

### Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

### Prompt:-i

Create a button that switches between light and dark themes using JavaScript by toggling CSS classes.

Ensure proper color contrast and accessibility so the theme changes instantly when the button is clicked

### Code:-

```
60  //-Task-2
61  // Create a button that switches between light and dark themes using JavaScript by toggling CSS classes.
62  Ensure proper color contrast and accessibility so the theme changes instantly when the button is clicked
63  <!DOCTYPE html>
64  <html lang="en">
65  <head>
66      <meta charset="UTF-8">
67      <meta name="viewport" content="width=device-width, initial-scale=1.0">
68      <title>Theme Switcher</title>
69      <style>
70          body {
71              font-family: Arial, sans-serif;
72              margin: 0;
73              padding: 20px;
74              transition: background-color 0.3s, color 0.3s;
75          }
76          .light-theme {
77              background-color: #ffffff;
78              color: #000000;
79          }
80          .dark-theme {
81              background-color: #333333;
82              color: #ffffff;
83          }
84          button {
85              padding: 10px 20px;
86              font-size: 16px;
87              cursor: pointer;
88          }
89      </style>
90  </head>
```

```

90 </head>
91 <body class="light-theme">
92   <h1>Welcome to the Theme Switcher</h1>
93   <p>Click the button below to toggle between light and dark themes.</p>
94   <button id="theme-toggle">Switch Theme</button>
95
96   <script>
97     const toggleButton = document.getElementById('theme-toggle');
98     const body = document.body;
99
100    toggleButton.addEventListener('click', () => {
101      body.classList.toggle('light-theme');
102      body.classList.toggle('dark-theme');
103    });
104  </script>
105 </body>
106
107 </html>

```

### Output:-

#### In white Theme

//Task-2 // Create a button that switches between light and dark themes using JavaScript by toggling CSS classes. Ensure proper color contrast and accessibility so the theme changes instantly when the button is clicked

#### Welcome to the Theme Switcher

Click the button below to toggle between light and dark themes.

Switch Theme

#### In Black Theme

# Welcome to the Theme Switcher

Click the button below to toggle between light and dark themes.

Switch Theme

### Justification:-

Implements a theme toggle button that quickly switches between light and dark modes using CSS classes.

Enhances accessibility and user experience by changing colors dynamically without refreshing the page.

### Task Description 3 – (User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

#### Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.

**Prompt:-**

Create a feedback form that collects a username, rating, and comments.

Use JavaScript to validate the required fields and ensure the form has a clear and user-friendly layout.

**Code:-**

```
109 //Task-3
110 //Create a feedback form that collects a username, rating, and comments.
111 Use JavaScript to validate the required fields and ensure the form has a clear and user-friendly layout.
112 <!DOCTYPE html>
113 <html lang="en">
114 <head>
115     <meta charset="UTF-8">
116     <meta name="viewport" content="width=device-width, initial-scale=1.0">
117     <title>Feedback Form</title>
118     <style>
119         body {
120             font-family: Arial, sans-serif;
121             margin: 0;
122             padding: 20px;
123         }
124         form {
125             max-width: 400px;
126             margin: auto;
127         }
128         label {
129             display: block;
130             margin-bottom: 5px;
131         }
132         input, textarea, select {
133             width: 100%;
```

```

132     input, textarea, select {
133         width: 100%;
134         padding: 10px;
135         margin-bottom: 15px;
136         border: 1px solid #ccc;
137         border-radius: 4px;
138     }
139     button {
140         padding: 10px 20px;
141         font-size: 16px;
142         cursor: pointer;
143     }
144     .error {
145         color: red;
146         margin-bottom: 15px;
147     }
148 </style>
149 </head>
150 <body>
151     <h1>Feedback Form</h1>
152     <form id="feedback-form">
153         <div class="error" id="error-message"></div>
154         <label for="username">Username:</label>
155         <input type="text" id="username" name="username" required>
156
157         <label for="rating">Rating:</label>
158         <select id="rating" name="rating" required>
159             <option value="">Select a rating</option>
160             <option value="1">1 - Poor</option>
161             <option value="2">2 - Fair</option>
162             <option value="3">3 - Good</option>
163             <option value="4">4 - Very Good</option>

```

```

158     <select id="rating" name="rating" required>
161         <option value="2">2 - Fair</option>
162         <option value="3">3 - Good</option>
163         <option value="4">4 - Very Good</option>
164         <option value="5">5 - Excellent</option>
165     </select>
166
167     <label for="comments">Comments:</label>
168     <textarea id="comments" name="comments" rows="4" required></textarea>
169
170     <button type="submit">Submit Feedback</button>
171 </form>
172
173 <script>
174     const form = document.getElementById('feedback-form');
175     const errorMessage = document.getElementById('error-message');
176
177     form.addEventListener('submit', (event) => {
178         event.preventDefault();
179         errorMessage.textContent = '';
180
181         const username = document.getElementById('username').value.trim();
182         const rating = document.getElementById('rating').value;
183         const comments = document.getElementById('comments').value.trim();
184
185         if (!username || !rating || !comments) {
186             errorMessage.textContent = 'Please fill in all fields.';
187             return;
188         }
189
190         alert('Thank you for your feedback!');
191         form.reset();
192     });
193 </script>
194 </body>
195 </html>

```

**Output:-**

Username:

Rating:

Select a rating 

Comments:



Submit Feedback

**Justifications:-**

Builds a feedback form capturing username, rating, and comments with JavaScript validation for required fields. Enhances usability by preventing incomplete submissions and guiding users to correct input errors.

**Task Description - 4 (Image Gallery with JavaScript)**

- **Task:** Display a small image gallery on a web page.

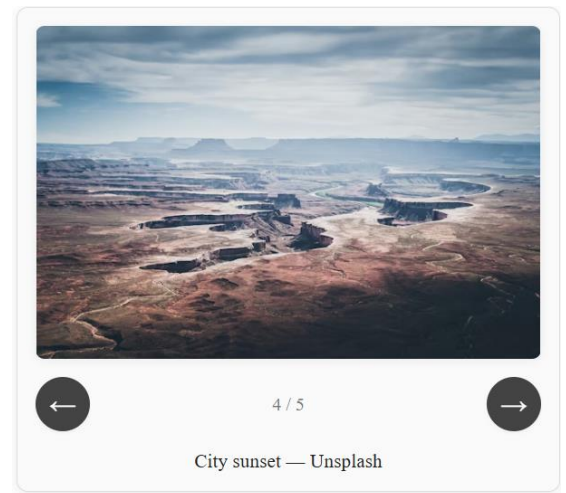
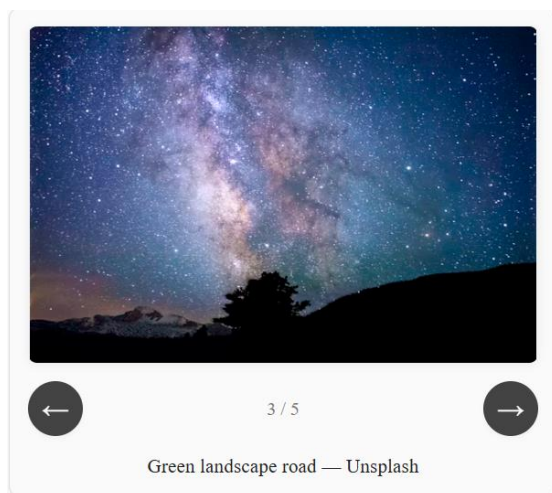
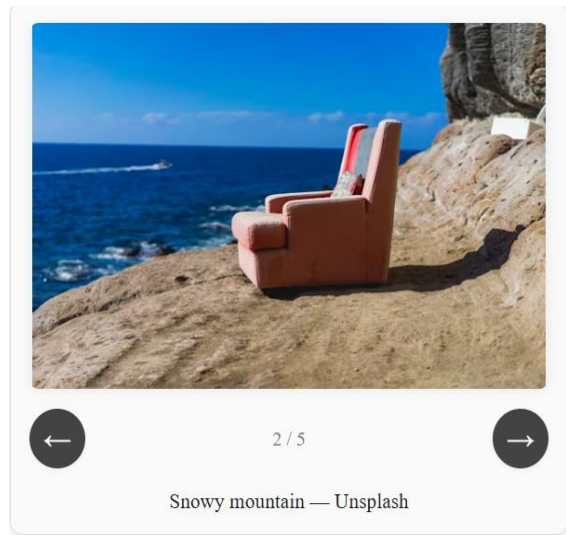
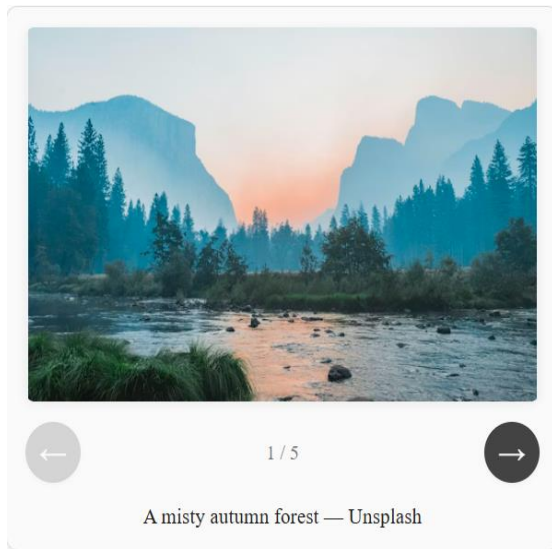
**Requirements:**

- Use JavaScript to dynamically load images.
- Add next/previous navigation buttons.
- Use AI to optimize layout responsiveness.

**Prompt:-**

create a small image gallery on a webpage using html,js to dynamically load images and add navigations on next/previous buttons and optimize layout responsiveness, HTML structure for the image gallery

**Output:-**



### Justification:-

Creates a responsive image gallery that dynamically loads images and supports next/previous navigation buttons. Provides a clean, interactive browsing experience optimized for different screen sizes

### Code:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Responsive JavaScript Image Gallery</title>
  <style>
    :root {
      --gallery-max-width: 480px;
      --gallery-margin: 0 auto;
      --gallery-bg: #fafafa;
      --gallery-border: 1px solid #ddd;
      --gallery-radius: 10px;
      --btn-bg: #333;
      --btn-color: #fff;
      --btn-radius: 50%;
      --btn-size: 48px;
    }

    .gallery-container {
      max-width: var(--gallery-max-width);
      margin: var(--gallery-margin);
      background: var(--gallery-bg);
      border: var(--gallery-border);
      border-radius: var(--gallery-radius);
      padding: 16px;
      box-sizing: border-box;
      display: flex;
      flex-direction: column;
      align-items: center;
      box-shadow: 0 2px 10px 0 rgba(0,0,0,.08);
    }

    .gallery-image-wrapper {
      position: relative;
      width: 100%;
      padding-top: 66%; /* Optimized for 3:2 ratio for
      background: #f1f2f4;
      border-radius: 8px;
      overflow: hidden;
      margin-bottom: 16px;
      transition: box-shadow .18s;
      box-shadow: 0 2px 8px 0 rgba(0,0,0,.06);
      display: flex;
      align-items: center;
      justify-content: center;
    }

    .gallery-image-wrapper img {
      margin: 0 auto;
      position: absolute;
      left: 0; top: 0; width: 100%; height: 100%;
      object-fit: contain;
      transition: opacity .16s;
      user-select: none;
      pointer-events: none;
    }
  
```

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```

    .gallery-buttons {
      display: flex;
      justify-content: space-between;
      align-items: center;
      margin-bottom: 8px;
    }

    .gallery-btn {
      background: var(--btn-bg);
      color: var(--btn-color);
      border: none;
      border-radius: var(--btn-radius);
      width: var(--btn-size);
      height: var(--btn-size);
      font-size: 1.5em;
      cursor: pointer;
      transition: background .18s, filter .18s;
      display: flex;
      align-items: center;
      justify-content: center;
      outline: none;
      opacity: 0.92;
      box-shadow: 0 2px 4px 0 #acacac22;
      z-index: 2;
    }

    .gallery-btn:disabled {
      background: #bbb;
      cursor: not-allowed;
      opacity: 0.6;
    }

    .gallery-caption {
      font-size: 1.07em;
      text-align: center;
      margin-top: 0.5em;
      color: #222;
    }

    @media (max-width: 600px) {
      .gallery-container {
        padding: 8px;
        max-width: 98vw;
      }

      .gallery-image-wrapper {
        padding-top: 60%;
        border-radius: 6px;
      }

      .gallery-btn {
        width: 40px;
        height: 40px;
        font-size: 1.1em;
      }

      .gallery-caption {
        font-size: 0.97em;
      }
    }

    @media (max-width: 400px) {
      .gallery-btn {
        width: 36px;
        height: 36px;
        font-size: 0.95em;
      }

      .gallery-image-wrapper {
        padding-top: 56%;
      }
    }
  
```

```

</style>
</head>
<body>
  <div class="gallery-container">
    <div class="gallery-image-wrapper" id="imgWrapper">
      <!-- The image will be inserted here by JS -->
    </div>
    <div class="gallery-buttons">
      <button class="gallery-btn" id="prevBtn" aria-label="Previous image">&larr;</button>
      <span id="galleryCounter" style="min-width: 50px;text-align: center;font-size: 1em; color: #777"></span>
      <button class="gallery-btn" id="nextBtn" aria-label="Next image">&rarr;</button>
    </div>
    <div class="gallery-caption" id="imgCaption"></div>
  </div>
<script>
  // AI optimized: Use small, modern placeholder pics & alt text for ally
  const images = [
    {
      src: "https://images.unsplash.com/photo-1506744038136-46273834b3fb?auto=format&fit=crop&w=600&q=80",
      alt: "A forest in autumn morning mist",
      caption: "A misty autumn forest – Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1519125323398-675f0ddb6308?auto=format&fit=crop&w=600&q=80",
      alt: "Snowy mountain peak with clouds",
      caption: "Snowy mountain – Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1465101046530-73398c7f28ca?auto=format&fit=crop&w=600&q=80",
      alt: "Winding road in green landscape",
      caption: "Green landscape road – Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1519985176271-adb1088fa94c?auto=format&fit=crop&w=600&q=80",
      alt: "City skyline at sunset",
      caption: "City sunset – Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1502082553048-f009c37129b9?auto=format&fit=crop&w=600&q=80",
      alt: "Closeup of sunflower",
      caption: "Sunflower closeup – Unsplash"
    }
  ];

  let currentIndex = 0;

  const imgWrapper = document.getElementById('imgWrapper');
  const prevBtn = document.getElementById('prevBtn');
  const nextBtn = document.getElementById('nextBtn');
  const imgCaption = document.getElementById('imgCaption');
  const galleryCounter = document.getElementById('galleryCounter');

```

```

function updateGallery(idx) {
  // Remove previous img
  imgWrapper.innerHTML = "";
  // Responsive: use srcset for layout and bandwidth optimization
  const img = document.createElement('img');
  img.src = images[idx].src;
  img.alt = images[idx].alt;
  img.decoding = "async";
  img.loading = "lazy";
  img.srcset = images[idx].src.replace("w=600", "w=400") + " 400w, " +
    images[idx].src + " 600w";
  img.sizes = "(max-width: 480px) 98vw, 480px";
  imgWrapper.appendChild(img);

  imgCaption.textContent = images[idx].caption;
  galleryCounter.textContent = (idx+1) + " / " + images.length;

  prevBtn.disabled = idx === 0;
  nextBtn.disabled = idx === images.length - 1;
}

// Handle navigation
prevBtn.addEventListener('click', () => {
  if (currentIndex > 0) {
    currentIndex--;
    updateGallery(currentIndex);
  }
});
nextBtn.addEventListener('click', () => {
  if (currentIndex < images.length - 1) {
    currentIndex++;
    updateGallery(currentIndex);
  }
});

// Keyboard accessibility
document.addEventListener('keydown', e => {
  if (e.key === "ArrowLeft" && currentIndex > 0) {
    currentIndex--;
    updateGallery(currentIndex);
  }
  if (e.key === "ArrowRight" && currentIndex < images.length - 1) {
    currentIndex++;
    updateGallery(currentIndex);
  }
});

// Initial load
updateGallery(currentIndex);
</script>
</body>
</html>

```

## Task Description - 5: (Notification Banner)

- Display a notification banner at the top of the page.

### Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

### Prompt:-

create a webpage to display the notification banner at the top of the page using html,js to show the hidden banner and style the banner using css to ensure smooth transition and clean UI behaviour.

### Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Notification Banner Example</title>
  <style>
    :root {
      --banner-bg: #3867d6;
      --banner-color: #fff;
      --banner-height: 48px;
      --banner-font: 1.05em;
      --banner-radius: 0 0 7px 7px;
      --banner-shadow: 0 2px 12px rgba(44,62,80,0.08);
      --banner-transition: transform 0.35s cubic-bezier(.8,.22,.23,.89), opacity .25s;
      --close-btn-size: 28px;
      --close-btn-bg: transparent;
      --close-btn-hover: #fff5;
      --close-btn-color: #fff;
    }
    body {
      margin: 0;
      font-family: system-ui,sans-serif;
      background: #f5f6fa;
      min-height: 400px;
    }
    .notif-banner {
      box-sizing: border-box;
      width: 100%;
      background: var(--banner-bg);
      color: var(--banner-color);
      min-height: var(--banner-height);
      display: flex;
      align-items: center;
      justify-content: space-between;
      position: fixed;
      left: 0;
      top: 0;
      z-index: 1000;
      border-radius: var(--banner-radius);
      box-shadow: var(--banner-shadow);
      padding: 0 24px 0 16px;
      font-size: var(--banner-font);
      letter-spacing: 0.01em;
      transform: translateY(-110%);
      opacity: 0;
      pointer-events: none;
      transition: var(--banner-transition);
    }
    .notif-banner.show {
      transform: translateY(0);
      opacity: 1;
      pointer-events: auto;
    }
    .notif-banner__text {
      flex: 1;
      padding: 6px 0;
      text-align: left;
    }
    .notif-banner__close {
      appearance: none;
      border: none;
      background: var(--close-btn-bg);
      color: var(--close-btn-color);
      border-radius: 50%;
      width: var(--close-btn-size);
      height: var(--close-btn-size);
      font-size: 1.22em;
      cursor: pointer;
      display: flex;
      align-items: center;
      justify-content: center;
      transition: background 0.18s;
      outline: none;
      margin-left: 14px;
    }
    .notif-banner__close:focus,
    .notif-banner__close:hover {
      background: var(--close-btn-hover);
    }
  </style>
</head>
<body>
```

```
    .notif-banner__text {
      flex: 1;
      padding: 6px 0;
      text-align: left;
    }
    .notif-banner__close {
      appearance: none;
      border: none;
      background: var(--close-btn-bg);
      color: var(--close-btn-color);
      border-radius: 50%;
      width: var(--close-btn-size);
      height: var(--close-btn-size);
      font-size: 1.22em;
      cursor: pointer;
      display: flex;
      align-items: center;
      justify-content: center;
      transition: background 0.18s;
      outline: none;
      margin-left: 14px;
    }
    .notif-banner__close:focus,
    .notif-banner__close:hover {
      background: var(--close-btn-hover);
    }
  </style>
  <script>
    /* Display 'Show' button only when banner is hidden */
    #showBannerBtn {
      position: fixed;
      top: 14px;
      right: 16px;
      z-index: 900;
      padding: 7px 16px;
      background: #fff;
      border: 1px solid #3867d6;
      color: #3867d6;
      border-radius: 18px;
      font-size: 1em;
      font-weight: 500;
      cursor: pointer;
      box-shadow: 0 0 8px #3867d610;
      opacity: 0;
      pointer-events: none;
      transition: opacity 0.22s;
    }
    #showBannerBtn.active {
      opacity: 1;
      pointer-events: auto;
    }
    @media (max-width: 480px) {
      .notif-banner {
        font-size: .95em;
        padding-right: 8px;
      }
      #showBannerBtn {
        font-size: .97em;
        right: 8px;
      }
    }
  </script>
</body>
```

```

<body>
  <div class="notif-banner" id="notifBanner" aria-live="polite" role="status">
    <span class="notif-banner__text">🔥 This is a notification banner. You can dismiss it with the close button!</span>
    <button class="notif-banner__close" id="closeBannerBtn" aria-label="Dismiss notification">&times;</button>
  </div>
  <button id="showBannerBtn">Show Notification</button>

  <script>
    // Elements
    const banner = document.getElementById('notifBanner');
    const closeBtn = document.getElementById('closeBannerBtn');
    const showBtn = document.getElementById('showBannerBtn');

    // Show the banner on page load (AI/UX: small delay to catch the eye)
    setTimeout(() => {
      banner.classList.add('show');
      showBtn.classList.remove('active');
    }, 250);

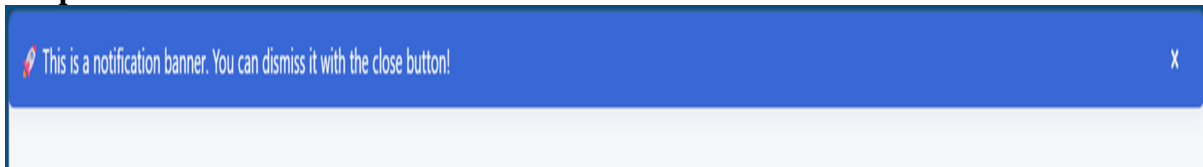
    // Hide the banner
    closeBtn.addEventListener('click', () => {
      banner.classList.remove('show');
      // Show the 'show' button for accessibility
      setTimeout(() => showBtn.classList.add('active'), 400);
    });

    // Show the banner again when user clicks 'Show Notification'
    showBtn.addEventListener('click', () => {
      banner.classList.add('show');
      showBtn.classList.remove('active');
    });

    // Optional: Hide banner on Escape key for ally
    document.addEventListener('keydown', e => {
      if (e.key === "Escape" && banner.classList.contains('show')) {
        banner.classList.remove('show');
        setTimeout(() => showBtn.classList.add('active'), 400);
      }
    });
  </script>
</body>
</html>

```

### Output:-



### Justification:-

Displays a notification banner at the top of the page that can be shown or hidden using JavaScript. Uses smooth transitions and styled layout to deliver important messages without disrupting the overall UI.